No Distance Matrix Required: Multivariate Regression as Supervised Metric Learning

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Abstract

Many applied machine learning problems need two things at once: accurate prediction of outcomes and a reusable notion of similarity for segmentation, clustering, and visualization. In practice, these are often built with separate tools, which can lead to segments that are easy to compute but weakly related to outcomes, or models that predict well but do not yield a stable distance function for downstream analytics. A common way to align geometry with supervision is through explicit Gram or distance matrix construction, which requires $O(n^2p)$ time and $O(n^2)$ storage, which becomes prohibitive at industry scale.

This paper shows how multivariate regression already provides both predictions and geometry from its fitted coefficient matrix. Let T_c and Y_c denote the column-centered target and feature matrices, respectively; let B be a linear map, and set embedding $Z := Y_cB$. Multivariate regression produces predictions $\widehat{T} = Y_cB$ in the usual way; we show that the same map B simultaneously induces a scalable, out-of-sample Mahalanobis-type distance through $M = BB^\top \succeq 0$, with no explicit distance matrix required or formed. The resulting framework, *Target Multidimensional Scaling* (TMDS), reverses the logic of classical MDS: rather than recovering coordinates from distances, it learns a single linear map from features to targets and extracts geometry directly from the fitted object B.

The paper makes three linked contributions that hold for any linear map from features to targets: (i) the observation that one regression coefficient matrix delivers both multivariate predictions as well as a corresponding low-rank PSD metric on features, (ii) a scalable out-of-sample embedding construction that avoids explicit $n \times n$ distance or Gram matrices, and (iii) a deterministic, assumption-free bound linking coordinate fit to centered Gram-matrix alignment.

Writing $\varepsilon := \|T_c - Z\|_F$ for the coordinate residual, the centered Gram mismatch satisfies the bound

$$\|T_cT_c^\top - ZZ^\top\|_F \;\le\; 2\,\|T_c\|_F\,\varepsilon + \varepsilon^2.$$

Multivariate regressions such as ridge (for stability and speed) therefore control geometry alignment through the coordinate objective $\varepsilon := \|T_c - Y_cB\|_F$. Euclidean distances in Z equal quadratic-form distances under M in feature space, so clustering in Z is exactly equivalent to clustering in Y under a supervised target metric. The entire construction requires only $O(np^2)$ cross-products and low-rank factors. Since no $n \times n$ matrix is formed at any stage, it becomes appealing in large n and moderate p regimes.

Keywords: Target multidimensional scaling; Supervised metric learning; Scalable clustering; Fast k-means; Low-rank matrix factorization; Mahalanobis distance learning

1 Introduction

Many applied machine learning problems need both prediction and geometry. One often wants a machine learning model that predicts outcomes accurately and, from the same learned object, a notion of similarity that can support segmentation, clustering, nearest-neighbor search, and visualization. In practice, these goals are commonly addressed with separate tools. A supervised model may predict well but not emit a reusable distance function, while an unsupervised embedding may provide convenient geometry that is only weakly related to the outcomes of interest. A common

way to align geometry with supervision is through explicit Gram or distance matrix construction, which requires $O(n^2p)$ time and $O(n^2)$ storage and becomes prohibitive at large scale.
This paper develops a simple, scalable estimator for that gap. The starting point is a feature matrix $Y \in \mathbb{R}^{n\times p}$ and a target matrix $T \in \mathbb{R}^{n\times q}$. The fitted object is the centered multivariate ridge map

$$\widehat{B} = \arg\min_{B\in\mathbb{R}^{p\times q}} \|T_c - Y_c B\|_F^2 + \lambda\|B\|_F^2, \qquad \lambda > 0,$$

where $Y_c := HY$, $T_c := HT$, and $H := I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$ is the centering matrix. Once the map $\widehat{B}$ is fit, it is used twice. First, it provides multivariate prediction. Second, it induces a quadratic-form geometry through

$$\widehat{M} := \widehat{B}\widehat{B}^\top \succeq 0,$$

and therefore an out-of-sample Mahalanobis-type distance together with a linear embedding. The minimizer is the standard multivariate ridge solution by design: the contribution is not a new estimator, but the observation that this familiar object is fast to fit and simultaneously induces a reusable out-of-sample geometry with a deterministic link to centered Gram alignment.
The paper is intentionally conservative about novelty claims. The building blocks are standard: classical MDS and strain matching,[1, 5, 16] Mahalanobis metric learning,[12, 13] multi-output ridge regression and reduced-rank approximations,[4, 7, 9, 14] and centered Gram similarity ideas.[2, 6, 10] The contribution here is the unified assembly of these components into a single out-of-sample geometry layer, together with a deterministic link between coordinate fit and centered Gram mismatch.

Primary contributions.

1. **A scalable estimator built from standard components.** A practical TMDS fit reduces to one multi-output ridge solve, optionally followed by rank truncation, and therefore avoids explicit $n \times n$ distance or Gram construction.

2. **TMDS as "reverse classical MDS" for supervised, out-of-sample geometry.** We formulate target-aligned geometry learning as centered Gram alignment in feature space and show how a single fitted linear map yields both multivariate, non-linear prediction and a reusable out-of-sample Mahalanobis-type distance.

3. **A deterministic diagnostic bound linking coordinate fit to geometry fit.** We prove an assumption-free inequality that upper-bounds the centered Gram mismatch in terms of the coordinate regression residual $\varepsilon = \|T_c - Y_c B\|_F$. Because ε is cheap to compute, the bound doubles as a practical diagnostic.

4. **A large-n usage regime.** A minimal genomics example shows that the estimator is useful precisely in the regime it targets: supervised, out-of-sample geometry at a scale where explicit all-pairs methods begin to be unattractive. Though the demonstration is linear, the scaling equally applies to non-linear methods.

TMDS is best understood as a feature-space method engineered to reproduce a distance-space geometric objective. Classical MDS starts from all-pairs dissimilarities and recovers coordinates through a centered Gram construction.[1, 5, 16] Mahalanobis metric learning learns a positive semidefinite matrix from supervision.[12] Multi-output ridge learns a stable linear map between two views.[7] TMDS sits at their intersection: the learned regression map is reinterpreted geometrically through the induced quadratic form and embedding.

2 From Classical MDS to TMDS

2.1 Classical multidimensional scaling (MDS)

Suppose δ_{ij} are observed dissimilarities. *Metric* MDS is often phrased as choosing an embedding $z_1, \ldots, z_n \in \mathbb{R}^r$ to match these distances by minimizing a stress criterion of the form[1]

$$\min_{\{z_i\}} \left(\sum_{i=1}^{n} \sum_{j<i} \left(\|z_i - z_j\|_2 - \delta_{ij} \right)^2 \right)^{1/2}.$$

In contrast, *classical* MDS [16] yields a closed-form construction when the dissimilarities correspond to squared Euclidean distances.

Let $D^{(2)} \in \mathbb{R}^{n \times n}$ be the matrix of squared dissimilarities, $D^{(2)}_{ij} = \delta^2_{ij}$. Classical MDS forms the double-centered matrix

$$G := -\frac{1}{2} H D^{(2)} H.$$

When the dissimilarities are squared Euclidean distances, i.e., there exist points $x_1, \ldots, x_n \in \mathbb{R}^r$ with $D^{(2)}_{ij} = \|x_i - x_j\|_2^2$, and writing $X \in \mathbb{R}^{n \times r}$ for the matrix with rows $x_i^\top$ and $X_c := HX$, we have

$$G = X_c X_c^\top,$$

so G is the centered Gram matrix of the unknown centered configuration.

Conversely, for any Gram matrix $G = XX^\top$, the induced squared Euclidean distances between rows satisfy

$$D^{(2)}_{ij} = G_{ii} + G_{jj} - 2G_{ij},$$

equivalently,

$$D^{(2)} = \operatorname{diag}(G)\mathbf{1}^\top + \mathbf{1} \operatorname{diag}(G)^\top - 2G.$$

Thus, when $D^{(2)}$ is squared Euclidean, the double-centering formula recovers the centered Gram matrix.

Since G is real and symmetric, it admits an eigendecomposition

$$G = V \Lambda V^\top.$$

An r-dimensional embedding is then obtained as

$$Z = V_r \Lambda_r^{1/2},$$

where V_r contains the eigenvectors associated with the largest positive eigenvalues, and Λ_r is the diagonal matrix of those eigenvalues. Negative eigenvalues indicate incompatibility with an exact Euclidean embedding or numerical noise and are typically discarded.

2.2 Reversing MDS

Target Multidimensional Scaling can be understood as reversing the input/output roles of classical MDS while keeping the same Gram-centric view of geometry. Classical MDS takes pairwise dissimilarities as primitive and returns an embedding via a centered Gram construction and an eigendecomposition. TMDS instead takes observed feature and target coordinates as primitive and learns a linear map B so that the centered Gram geometry induced by features matches the centered Gram geometry of the targets.

Visual summary.

$$\textbf{Classical MDS:}\quad \underbrace{\{\delta_{ij}\}_{i<j}}_{\text{distances}} \longrightarrow \underbrace{D^{(2)}}_{\text{squared distances}} \xrightarrow{G=-\frac{1}{2}H(\cdot)H} \underbrace{G}_{\text{Gram}} \xrightarrow{\text{eigendecomp.}} \underbrace{Z}_{\text{coordinates}}$$

$$\textbf{TMDS:}\quad \underbrace{(Y,T)}_{\text{coordinates}} \longrightarrow \underbrace{\text{learn } B \text{ (and } M = BB^\top)}_{\text{operator/metric}} \longrightarrow \underbrace{Z := Y_c B}_{\text{coordinates}} \Longrightarrow \underbrace{ZZ^\top}_{\text{Gram}} \approx \underbrace{T_c T_c^\top}_{\text{target Gram}} .$$

In both cases, the centered Gram matrix is the central geometric object. Classical MDS constructs it from distances and then recovers coordinates, whereas TMDS learns a linear operator so that the feature-induced centered Gram matches target geometry.

3 Target Multidimensional Scaling: Centered Gram Alignment

3.1 Setup and goal

Let the data comprise

$Y \in \mathbb{R}^{n\times p}$ (features),

$T \in \mathbb{R}^{n\times q}$ (targets: continuous coordinates, one-hot class indicators, or mixtures thereof).

The goal of TMDS is to learn a feature-space matrix $M \in \mathbb{R}^{p\times p}$ such that the geometry induced by Y under M matches the geometry of T. We allow $M \succeq 0$, so the induced distance may be a pseudometric rather than a strict metric.

Assume M is symmetric positive semidefinite. Then M admits a factorization $M = BB^\top$ and induces the Mahalanobis-type distance[12, 13]

$$d_M(y_i, y_j) = \sqrt{(y_i - y_j)^\top M (y_i - y_j)}.$$

If $M \succ 0$, this is a metric on $\mathbb{R}^p$; if $M \succeq 0$, it is in general only a pseudometric because $d_M(y_i, y_j) = 0$ can occur for distinct points whose difference lies in $\mathrm{null}(M)$.

3.2 Gram alignment

To compare geometry rather than absolute location, it is natural to compare centered Gram matrices, since for Euclidean point configurations the centered Gram matrix encodes the same information as the pairwise squared distances up to translation[1, 5]. Let

$$Y_c := HY, \qquad T_c := HT.$$

The centered target Gram matrix is

$$G_T := T_c T_c^\top,$$

and the centered feature Gram matrix induced by M is

$$G_Y(M) := Y_c M Y_c^\top.$$

TMDS in its most geometry-faithful form seeks $M \succeq 0$ such that

$$G_T \approx G_Y(M),$$

for example by minimizing

$$\min_{M \succeq 0} \left\| T_c T_c^\top - Y_c M Y_c^\top \right\|_F^2 .$$

Unlike stress-based formulations in metric and non-metric MDS,[1, 11] this poses TMDS as centered Gram (strain) alignment, closely analogous to classical MDS.[1]

3.3 Low-rank factorization and an explicit embedding

Assume $M \succeq 0$. Then there exists a factorization

$$M = BB^\top \qquad \text{for some } B \in \mathbb{R}^{p \times r},$$

where r is an embedding dimension. Substituting $M = BB^\top$ into the centered feature Gram gives

$$G_Y(M) = Y_c B B^\top Y_c^\top = (Y_c B)(Y_c B)^\top .$$

Define the learned embedding

$$Z := Y_c B \in \mathbb{R}^{n \times r}.$$

Thus, aligning centered Gram matrices is equivalent to seeking a linear map B such that the centered embedding $Z = Y_c B$ reproduces the centered target Gram:

$$T_c T_c^\top \approx (Y_c B)(Y_c B)^\top .$$

Euclidean distances in the embedding correspond exactly to quadratic-form distances in feature space:

$$\|z_i - z_j\|_2^2 = \left\| (y_i - y_j)^\top B \right\|_2^2 = (y_i - y_j)^\top B B^\top (y_i - y_j) = (y_i - y_j)^\top M (y_i - y_j).$$

This identity is unaffected by centering because differences $y_i - y_j$ are translation-invariant.

3.4 Non-uniqueness under Gram alignment

Gram matrices, and therefore pairwise Euclidean distances, are invariant under right-multiplication by an orthogonal matrix. If $Q \in \mathbb{R}^{q \times q}$ satisfies $Q^\top Q = I$, then

$$(T_c Q)(T_c Q)^\top = T_c T_c^\top .$$

Therefore, Gram alignment identifies geometry up to an orthogonal change of coordinates; it does not, by itself, pick a unique coordinate representative of T. When a specific coordinate representative is required for interpretation, orthogonal Procrustes provides a standard post-processing step.[15]

4 A bounded, scalable surrogate for Gram alignment

4.1 Assumption-free error bound on full Gram alignment

A useful link between coordinate error and Gram error is the submultiplicative inequality

$$\|JJ^\top - KK^\top\|_F \le (\|J\|_F + \|K\|_F)\, \|J - K\|_F,$$

which follows from the decomposition $JJ^\top - KK^\top = J(J-K)^\top + (J-K)K^\top$ and the bound $\|PQ^\top\|_F \le \|P\|_F\|Q\|_F$.[8] Applying this with $J = T_c$ and $K = Y_cB$ gives

$$\|T_cT_c^\top - (Y_cB)(Y_cB)^\top\|_F \le (\|T_c\|_F + \|Y_cB\|_F)\,\|T_c - Y_cB\|_F.$$

A slightly sharper assumption-free corollary follows by bounding $\|Y_cB\|_F$ in terms of T_c and the coordinate error. Define $Z := Y_cB$, its Gram $G_Z := ZZ^\top$, and $\varepsilon := \|T_c - Z\|_F$. Since $\|Z\|_F \le \|T_c\|_F + \varepsilon$, we obtain

$$\|G_T - G_Z\|_F = \|T_cT_c^\top - ZZ^\top\|_F \le \big(2\|T_c\|_F + \varepsilon\big)\varepsilon = 2\|T_c\|_F\,\varepsilon + \varepsilon^2. \tag{1}$$

Thus centered Gram mismatch is first-order in the coordinate error when $\varepsilon \ll \|T_c\|_F$. Appendix E.1 records a sharpened variant that replaces $\|T_c\|_F$ by the smaller operator norm $\|T_c\|_{\mathrm{op}}$, with no change to the estimator. The operator-norm bound in (4) can be tightened further by exploiting the structure of the residual relative to the target subspace; see Appendix E.2. All of these are deterministic upper bounds that suffice to certify small centered Gram mismatch, though none is necessary for Gram alignment.

4.2 Approximate Gram alignment using ridge

A natural and practical step is therefore to fit a specific coordinate alignment by classical multivariate regression, with ridge regression being a scalable and stable choice:[7]

$$\min_{B\in\mathbb{R}^{p\times q}} \ \|T_c - Y_cB\|_F^2 + \lambda\,\|B\|_F^2\,, \qquad \lambda \ge 0.$$

This objective is stronger than Gram alignment because it matches the target coordinates after centering, not only the induced Gram matrix. It fixes a particular coordinate system for T; translation is handled either by centering or, equivalently, by including an unpenalized intercept (Appendix B).

The same formulation also covers categorical and mixed targets. When T consists of one-hot class indicators, TMDS becomes a multivariate least-squares discriminant map rather than a continuous-response regressor. In that setting it is closely related to linear discriminant analysis[3] (LDA): under the usual equal within-class covariance assumptions, its score space can be viewed as an LDA-type discriminant subspace up to centering, scaling, and rotation. With ridge regularization and optional rank truncation, however, TMDS is best interpreted as a regularized low-rank discriminant analogue rather than an exact reformulation of classical LDA. More generally, one may concatenate standardized continuous target coordinates with appropriately weighted indicator columns; the learned geometry then reflects the chosen trade-off between regression fit and class separation.

For $\lambda > 0$, the minimizer has the closed form

$$\widehat{B} = \Big(Y_c^\top Y_c + \lambda I_p\Big)^{-1} Y_c^\top T_c.$$

The induced matrix is then $\widehat{M} = \widehat{B}\widehat{B}^\top \succeq 0$, and the fitted centered targets are

$$\widehat{T}_c := Y_c\widehat{B}.$$

For geometry, one may optionally compress $\widehat{B}$ to rank m via a post-hoc singular value truncation and then use the corresponding m-dimensional embedding.

4.3 Linearity, modularity, and basis design

The linear form of TMDS has at least two direct practical consequences. First, at the level of the full metric $\widehat{M} = \widehat{B}\widehat{B}^\top$, linearity yields an exact modular composition property: target columns can be added, removed, or reweighted through rank-one updates, with no discrepancy relative to joint fitting on the modified target set; see Appendix F. Second, TMDS inherits the limitations of linear fitting in the chosen basis. Nonlinear relationships are captured only to the extent that the original features are expanded into a richer representation $\Phi(Y)$.

TMDS is therefore best viewed as a downstream geometry layer. Nonlinearity is handled upstream through the chosen feature basis or representation, while TMDS turns that representation into both prediction and geometry through a stable linear solve. This separation of concerns is by design: it keeps the downstream estimator simple, interpretable, and scalable, but it also means that the attainable quality of geometry alignment is controlled by the expressivity of the upstream basis.

Computationally, supplementing the original features with additional nonlinear basis functions trades improved expressivity against a larger feature dimension p. Computing the cross-products $Y_c^\top Y_c$ and $Y_c^\top T_c$ costs $O(np^2)$ and $O(npq)$, respectively, while a Cholesky factorization of the $p \times p$ matrix $Y_c^\top Y_c + \lambda I_p$ costs $O(p^3)$. This is attractive when $n \gg p$, but it becomes costly when p is very large. In the $p \gg n$ regime, iterative least-squares solvers or dual formulations may be preferable.

5 Bounds on Gram alignment

5.1 Strict bounds on Gram matching with the scalable quadratic surrogate

It is useful to view the ridge surrogate as a tractable approximation to a more direct geometry-matching goal. Recall $Z := Y_c B$ and its centered Gram $G_Z := ZZ^\top$. A natural full TMDS geometry objective is

$$\min_{B:\ \mathrm{rank}(B) \le m} \|G_T - G_Z\|_F^2. \tag{2}$$

This is equivalent, up to the classical double-centering map, to an all-pairs squared-distance matching problem. Let $D_Z^{(2)}$ be the matrix with (i,j) entry $\|z_i - z_j\|_2^2$, and define $D_T^{(2)}$ analogously for the rows of T_c. From classical MDS theory,[1] $G_Z = -\frac{1}{2} H D_Z^{(2)} H$ and $G_T = -\frac{1}{2} H D_T^{(2)} H$, which gives

$$\|G_T - G_Z\|_F^2 = \tfrac{1}{4} \|H(D_T^{(2)} - D_Z^{(2)})H\|_F^2.$$

Thus (2) can be read as an all-pairs squared-distance matching problem rather than as a coordinate regression problem.

The advantage of the ridge surrogate is that it remains quadratic in B, avoids explicit $n \times n$ objects, and still controls geometry by (1). In that sense, it preserves the most useful part of the full Gram story while remaining solver-friendly.

5.2 Normalized form and connection to R-squared

Let $\widehat{T}_c := Y_c \widehat{B}$ and define $\varepsilon := \|T_c - \widehat{T}_c\|_F$. The total variance-weighted coefficient of determination is

$$R^2_{\text{total}} := 1 - \frac{\|T_c - \widehat{T}_c\|_F^2}{\|T_c\|_F^2} = 1 - \frac{\varepsilon^2}{\|T_c\|_F^2}.$$

Define $\rho := \varepsilon/\|T_c\|_F = \sqrt{1 - R^2_{\text{total}}}$. Then (1) becomes

$$\|G_T - G_Z\|_F \le \|T_c\|_F^2 \,(2\rho + \rho^2) = \|T_c\|_F^2 \left(2\sqrt{1 - R^2_{\text{total}}} + (1 - R^2_{\text{total}})\right).$$

Therefore, small coordinate error, equivalently high R^2_{total}, implies small centered Gram mismatch. This is the main reason the simple ridge surrogate is meaningful geometrically.

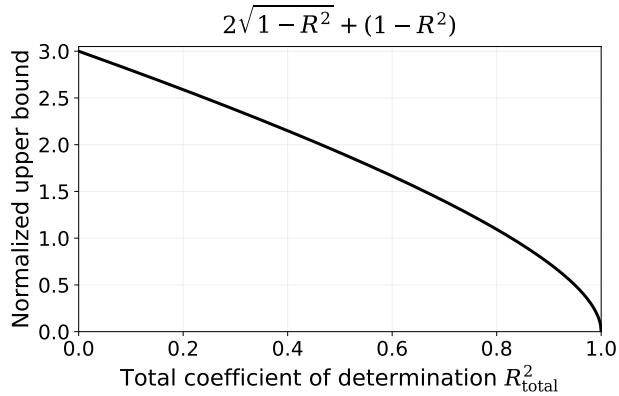

Figure 1: Normalized upper bound on centered Gram mismatch as a function of total coefficient of determination R^2_{total}. Normalizing the bound gives $\|G_T - G_Z\|_F/\|T_c\|_F^2 \le 2\sqrt{1 - R^2_{\text{total}}} + (1 - R^2_{\text{total}})$. The curve decreases monotonically to zero as $R^2_{\text{total}} \to 1$, implying that the normalized centered Gram mismatch is itself driven to zero in the limit of exact coordinate fit.

Exact coordinate fit implies exact centered Gram alignment:

$$Y_c\widehat{B} = T_c \quad \Longrightarrow \quad (Y_c\widehat{B})(Y_c\widehat{B})^\top = T_c T_c^\top.$$

The converse need not hold, because Gram alignment is invariant under orthogonal changes of coordinates. The implication from exact coordinate fit to exact centered Gram alignment is immediate from the definition of a Gram matrix. The useful point here is that the same relationship extends quantitatively: approximate coordinate fit controls centered Gram mismatch through the deterministic bound above.

5.3 Rank control via post-hoc truncation

Let $\widehat{B} = U\Sigma V^\top$ be a singular value decomposition and define the rank-m truncation

$$\widehat{B}_{(m)} := U_m \Sigma_m V_m^\top.$$

For geometry, it is convenient to use the reduced factor

$$\widetilde{B}_{(m)} := U_m \Sigma_m \in \mathbb{R}^{p\times m}, \qquad Z_{(m)} := Y_c \widetilde{B}_{(m)} \in \mathbb{R}^{n\times m},$$

with induced metric

$$\widetilde{M}_{(m)} := \widetilde{B}_{(m)} \widetilde{B}_{(m)}^\top.$$

Then for any pair i, j,

$$\|z_{(m),i} - z_{(m),j}\|_2^2 = (y_{c,i} - y_{c,j})^\top \widetilde{M}_{(m)} (y_{c,i} - y_{c,j}),$$

so Euclidean distances in the truncated embedding coincide exactly with Mahalanobis distances under $\widetilde{M}_{(m)}$. Appendix C gives predictive and metric perturbation bounds induced by the discarded singular spectrum.

6 Practical computation

6.1 Basic algorithmic recipe

TMDS estimates a linear map $\widehat{B}$ that can be used to build the quadratic-form distance induced by $\widehat{M} = \widehat{B}\widehat{B}^\top$ and to define an embedding $Z = Y_c\widehat{B}$ for clustering. In doing so, it also provides a least-squares estimator of $T \approx \widehat{T}$, where

$$\widehat{T} = \mathbf{1}\widehat{a}^\top + Y\widehat{B} = \mathbf{1}\bar{t}^\top + Y_c\widehat{B} = \mathbf{1}\bar{t}^\top + Z, \qquad \widehat{a}^\top := \bar{t}^\top - \bar{y}^\top\widehat{B}.$$

A minimal train step is:

1. Center Y and T to obtain Y_c and T_c.
2. Choose $\lambda > 0$ and solve
$$\left(Y_c^\top Y_c + \lambda I_p\right)\widehat{B} = Y_c^\top T_c.$$
3. Optionally compute a rank-m truncation $\widehat{B}_{(m)}$ and the associated geometry factor $\widetilde{B}_{(m)}$.
4. Use $\widehat{B}$ or $\widehat{B}_{(m)}$ for prediction, and $\widetilde{B}_{(m)}$ for geometry.

6.2 Computation and scaling

Let

$$A := Y_c^\top Y_c + \lambda I_p \in \mathbb{R}^{p\times p}, \qquad R := Y_c^\top T_c \in \mathbb{R}^{p\times q}.$$

For $\lambda > 0$, A is symmetric positive definite, so one can compute a Cholesky factorization

$$A = LL^\top$$

and solve $LL^\top\widehat{B} = R$ by two triangular solves. The Cholesky factorization is computed once and reused across all q target columns.

The dominant computational costs are:

- $O(np^2)$ to form $Y_c^\top Y_c$,
- $O(npq)$ to form $Y_c^\top T_c$,
- $O(p^3)$ for the Cholesky factorization of $Y_c^\top Y_c + \lambda I_p$,
- $O(p^2q)$ for the subsequent triangular solves,
- optionally $O(pqm)$ for a rank-m truncation of $\widehat{B}$.

This avoids constructing any $n \times n$ distance or Gram matrix and is therefore much cheaper than all-pairs distance-based methods when n is large.

When $n \ll p$, a dual (Woodbury) form can be cheaper:

$$\widehat{B} = Y_c^\top\left(Y_cY_c^\top + \lambda I_n\right)^{-1}T_c, \tag{3}$$

which replaces a $p \times p$ solve with an $n \times n$ solve.[4] When p is very large, iterative Krylov solvers are also attractive because they require only matrix-vector products with Y_c and $Y_c^\top$.

6.3 Orthogonal invariance of the induced Mahalanobis matrix

Let $Q \in \mathbb{R}^{q\times q}$ be orthogonal. If $\widehat{B}$ minimizes $\|T_c - Y_c B\|_F^2 + \lambda\|B\|_F^2$, then $\widehat{B}Q$ minimizes $\|T_c Q - Y_c B\|_F^2 + \lambda\|B\|_F^2$. Consequently,

$$(\widehat{B}Q)(\widehat{B}Q)^\top = \widehat{B}\widehat{B}^\top,$$

so while $\widehat{B}$ depends on the chosen coordinate basis of T, the learned matrix $\widehat{M} = \widehat{B}\widehat{B}^\top$ is invariant under orthogonal changes of target coordinates.

7 Minimal Scaling Example: label-aligned clustering on PBMC 68k genomics benchmark

7.1 Context

Single-cell RNA-seq genomics data can contain millions of cells, and the practical challenge is to recover meaningful cell populations without losing rare types as the dataset grows. The benchmark here is therefore about scaling and holdout performance together, not merely clustering the training set well. This is also the regime in which TMDS is most naturally motivated: one wants a supervised, out-of-sample geometry layer that can be fit without explicit all-pairs distances.

7.2 Setup: fixed holdout protocol

We benchmark the Zheng 2017 PBMC 68k dataset[17] using the FACS-derived cell-type labels as external reference classes. The subsampled labeled dataset available from Scanpy is split into a fixed training pool of 60,000 cells and a fixed holdout set of 5,877 cells, with the holdout never used for fitting. Every method uses the same counts-to-PCA preprocessing recipe fit on the current training subset only: library-size normalization, log transform, highly variable gene selection, and PCA to 250 components. Reported values are averaged over 10 iteration runs with fixed seeds.

7.3 Methods and supervision

We compare three inductive evaluation heads on the shared preprocessing pipeline: (i) direct MiniBatchKMeans on PCA features as the baseline, (ii) TMDS followed by MiniBatchKMeans, and (iii) the direct TMDS prediction head. For scaling reference, we also report a train-only Leiden graph-clustering baseline obtained by fitting Leiden on the training set and assigning holdout points by 1-nearest-neighbor label transfer.

TMDS uses requested embedding dimension $m = 250$, with effective rank capped by target width. Ridge regularization is $\lambda = 0.1$. The point of the example is not to claim dominance over all specialized single-cell methods, but to show that the estimator remains cheap to fit while preserving a supervised geometry that transfers to unseen cells. The Leiden values are included only as a train-only graph-clustering reference; because Leiden does not provide a native out-of-sample map, its holdout results should not be interpreted as those of an inductive clustering method in the same sense as TMDS or MiniBatchKMeans.

7.4 Scaling summary

At the largest training size $n = 60{,}000$, the direct TMDS prediction head gives holdout ARI/NMI 0.2696/0.3990. The corresponding values are 0.1627/0.3125 for TMDS followed by MiniBatchKMeans, 0.0301/0.0613 for direct MiniBatchKMeans, and 0.0021/0.0139 for Leiden transfer. At

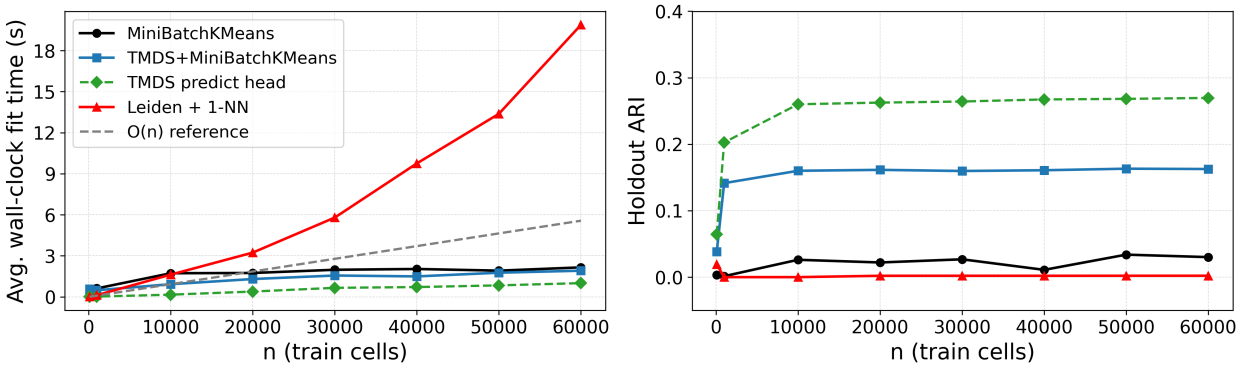

Figure 2: Left: fit time vs. training size with an $O(n)$ reference anchored at $n = 10{,}000$. Right: holdout ARI vs. training size for TMDS+MiniBatchKMeans, the direct TMDS prediction head, the MiniBatchKMeans baseline, and the train-only Leiden transfer reference.

the same training size, fit times are 1.01 s for TMDS prediction, 1.91 s for TMDS followed by MiniBatchKMeans, 2.16 s for MiniBatchKMeans, and 19.88 s for Leiden transfer.

7.5 Quality interpretation

Figure 2 is not intended to position TMDS as a complete solution to single-cell clustering. Its purpose is narrower: to verify that the estimator fulfills its design objectives. The model is lightweight to fit, the learned embedding generalizes to held-out data, and the supervised geometry yields markedly higher label coherence than unsupervised MiniBatchKMeans run on the same preprocessing pipeline. This is precisely the large-n regime in which avoiding explicit $n \times n$ constructions becomes essential.

8 Conclusion

8.1 Non-Linear Capacity

TMDS is a downstream *linear* geometry step. It can only align geometry to the extent that the centered targets T_c are well approximated by a linear map of the centered basis features $\Phi(Y)_c$. Thus the upstream feature basis effectively controls the attainable alignment: richer bases make $T_c \approx \Phi(Y)_c B$ more accurate, after which TMDS can compress the resulting map to rank m and cluster in a low-dimensional embedding.

8.2 Closing Remarks

The paper does not seek to replace specialized nonlinear representation learning, stress-based embedding, or pair/triplet metric learning. Its contribution is narrower: it shows that a standard multi-output ridge fit can be reinterpreted as an out-of-sample supervised geometry layer, that coordinate fit directly controls centered Gram mismatch through a simple deterministic bound, and that the resulting estimator inherits the favorable computational scaling of convex quadratic optimization.

8.3 Summary

This paper has assembled a practical link between prediction and geometry learning in a single reusable fitted object. TMDS fits a multivariate ridge map B once, then uses it twice:

1. once so that $Y \mapsto \widehat{T}$ provides multi-target, out-of-sample prediction via $\widehat{T} = \mathbf{1}\bar{t}^\top + Y_c\widehat{B}$,
2. a second time when $\widehat{M} = \widehat{B}\widehat{B}^\top \succeq 0$ provides a Mahalanobis-type distance and an embedding for scalable similarity, clustering, and visualization.

The main technical link is the assumption-free bounds in (1) and Appendix E, which turn coordinate fit into centered Gram control. The main practical consequence is that one can tune a simple, scalable estimator for predictive fit while retaining an explicit statement about geometric fidelity.

Appendices

Appendix A Basic vocabulary used throughout

We use the following terms consistently.

- **Geometry.** The structure that determines which observations are close, which are far, and which are neighbors.
- **Distance and similarity.** A geometry can be represented by distances or by inner-product similarities. In Euclidean settings, these are two views of the same information.[1]
- **Embedding.** An embedding is a new coordinate system for the observations, designed so that ordinary Euclidean distances in the embedding reflect relationships of interest in the original data.
- **Supervision.** Outcome information that reshapes geometry so that nearby points tend to have similar targets.
- **In-sample and out-of-sample.** In-sample geometry is defined only on a fixed training set. Out-of-sample geometry comes with an explicit rule that can be applied to new observations without recomputing all-pairs quantities.
- **Strain and stress.** Strain objectives match Gram structure derived from squared distances. Stress objectives match distances directly.[1, 11]

Appendix B TMDS Algorithm

Table 1: TMDS practical algorithm (train/predict), avoiding $n \times n$ distance or Gram construction.

Step	Operation
1	**Center training data.** Compute $\bar{y}^\top = \frac{1}{n}\mathbf{1}^\top Y$, $\bar{t}^\top = \frac{1}{n}\mathbf{1}^\top T$, then $Y_c := Y - \mathbf{1}\bar{y}^\top$, $T_c := T - \mathbf{1}\bar{t}^\top$.
2	**Form cross-products.** Compute $A := Y_c^\top Y_c + \lambda I_p$ and $R := Y_c^\top T_c$.
3	**Solve for $\widehat{B}$.** Solve $A\widehat{B} = R$ using a Cholesky factorization or an iterative solver.
4	**Optional rank control.** Compute a truncated SVD $\widehat{B} \approx U_m \Sigma_m V_m^\top$ and set $\widehat{B}_{(m)} := U_m \Sigma_m V_m^\top$, $\widetilde{B}_{(m)} := U_m \Sigma_m$.
5	**Recover intercept and fitted targets.** Set $\widehat{a}^\top := \bar{t}^\top - \bar{y}^\top \widehat{B}_{(m)}$, then $\widehat{T} = \mathbf{1}\widehat{a}^\top + Y\widehat{B}_{(m)}$.
6	**Predict and embed new data.** For Y_{new}, center with training means and compute $\widehat{T}_{\text{new}}$ and, if desired, $Z_{\text{new},(m)} := Y_{\text{new},c}\widetilde{B}_{(m)}$.

Appendix C Why centering can be dropped for ridge: equivalence via an intercept

Explicit centering can be avoided in implementation if one includes an intercept and does *not* penalize it. Consider

$$\min_{a\in\mathbb{R}^q,\, B\in\mathbb{R}^{p\times q}} \left\|T - \mathbf{1}a^\top - YB\right\|_F^2 + \lambda \|B\|_F^2 .$$

For fixed B, the minimizing intercept satisfies

$$a^\star(B)^\top = \frac{1}{n}\mathbf{1}^\top(T - YB) = \bar{t}^\top - \bar{y}^\top B.$$

Substituting $a^\star(B)$ back into the residual yields

$$T - \mathbf{1}a^\star(B)^\top - YB = (T - \mathbf{1}\bar{t}^\top) - (Y - \mathbf{1}\bar{y}^\top)B = HT - HYB.$$

Hence,

$$\min_{a,B} \left\|T - \mathbf{1}a^\top - YB\right\|_F^2 + \lambda \|B\|_F^2 \equiv \min_B \|HT - HYB\|_F^2 + \lambda \|B\|_F^2 .$$

This is the precise reason centering may be omitted when moving to ridge regression: the unpenalized intercept absorbs translations, which is exactly what centering removes.

Appendix D Compressed map $\widehat{B}$

Let $\widehat{B} = U\Sigma V^\top$ be the singular value decomposition, with singular values $\sigma_1 \geq \cdots \geq \sigma_r \geq 0$. The rank-$m$ truncation $\widehat{B}_{(m)} := U_m\Sigma_m V_m^\top$ retains

$$\|\widehat{B}_{(m)}\|_F^2 = \sum_{k\leq m} \sigma_k^2$$

and discards energy

$$\|\widehat{B} - \widehat{B}_{(m)}\|_F^2 = \sum_{k>m} \sigma_k^2.$$

For metric construction, the reduced factor $\widetilde{B}_{(m)} := U_m\Sigma_m$ induces the same rank-m metric:

$$\widehat{B}_{(m)}\widehat{B}_{(m)}^\top = \widetilde{B}_{(m)}\widetilde{B}_{(m)}^\top.$$

Let $X_\Phi \in \mathbb{R}^{n\times p}$ denote the centered feature matrix used to fit $\widehat{B}$ (e.g., $X_\Phi = \Phi(Y)_c$). Define the uncompressed and compressed fitted targets

$$\widehat{T} := X_\Phi\widehat{B}, \qquad \widehat{T}_{(m)} := X_\Phi\widehat{B}_{(m)}.$$

With $E := \widehat{B} - \widehat{B}_{(m)}$, the truncation perturbation satisfies

$$\|\widehat{T} - \widehat{T}_{(m)}\|_F = \|X_\Phi E\|_F \leq \|X_\Phi\|_F \, \|E\|_{\mathrm{op}} = \|X_\Phi\|_F \, \sigma_{m+1}.$$

A complementary bound is

$$\frac{1}{n}\|\widehat{T} - \widehat{T}_{(m)}\|_F^2 \leq \|\widehat{\Sigma}_\Phi\|_{\mathrm{op}} \sum_{k>m} \sigma_k^2, \qquad \widehat{\Sigma}_\Phi := \frac{1}{n}X_\Phi^\top X_\Phi.$$

Thus a fixed small m is justified precisely when the singular-value tail of $\widehat{B}$ decays rapidly.

Appendix E Additional bounds

Gram continuity (Frobenius norm). For any $J, K \in \mathbb{R}^{n\times r}$,

$$\|JJ^\top - KK^\top\|_F \le (\|J\|_F + \|K\|_F)\,\|J - K\|_F.$$

E.1 Operator-norm refinement of the main bound

The Frobenius-norm bound in (1) can be sharpened without changing the estimator. Let

$$E := T_c - Z = T_c - Y_c B.$$

Then

$$T_c T_c^\top - ZZ^\top = T_c E^\top + E T_c^\top - EE^\top.$$

Using $\|AB^\top\|_F \le \|A\|_{\mathrm{op}}\|B\|_F$ and $\|EE^\top\|_F \le \|E\|_F^2$, we obtain

$$\|T_c T_c^\top - ZZ^\top\|_F \le 2\|T_c\|_{\mathrm{op}}\,\|E\|_F + \|E\|_F^2. \tag{4}$$

Writing $\varepsilon := \|E\|_F = \|T_c - Y_c B\|_F$, this becomes

$$\|G_T - G_Z\|_F \le 2\|T_c\|_{\mathrm{op}}\,\varepsilon + \varepsilon^2.$$

Since $\|T_c\|_{\mathrm{op}} \le \|T_c\|_F$, (4) is never weaker than (1) and can be substantially tighter when the target columns are redundant or strongly correlated.

If we also define the ridge objective

$$F_\lambda(B) := \|T_c - Y_c B\|_F^2 + \lambda\|B\|_F^2,$$

then $\varepsilon^2 \le F_\lambda(B)$, so

$$\|G_T - G_Z\|_F \le 2\|T_c\|_{\mathrm{op}}\sqrt{F_\lambda(B)} + F_\lambda(B). \tag{5}$$

Thus the same convex loss that defines ridge regression directly controls a sharper centered-Gram bound.

Fixed target transforms. The same argument applies after any fixed linear target transform. Let $\widetilde{T}_c := T_c L$ for a chosen matrix $L \in \mathbb{R}^{q\times q'}$, let $\widetilde{Z} := Y_c\widetilde{B}$, and define $\widetilde{\varepsilon} := \|\widetilde{T}_c - \widetilde{Z}\|_F$. Then

$$\|\widetilde{T}_c\widetilde{T}_c^\top - \widetilde{Z}\widetilde{Z}^\top\|_F \le 2\|\widetilde{T}_c\|_{\mathrm{op}}\,\widetilde{\varepsilon} + \widetilde{\varepsilon}^2.$$

In particular, if $L = W^{1/2}$ for some $W \succeq 0$, then

$$\|T_c W T_c^\top - \widetilde{Z}\widetilde{Z}^\top\|_F \le 2\|T_c W^{1/2}\|_{\mathrm{op}}\,\widetilde{\varepsilon} + \widetilde{\varepsilon}^2.$$

This covers target standardization, importance weighting, whitening, or low-rank target compression as special cases. Such transforms can shrink the leading singular value of the fitted target and therefore tighten the constant in the bound, without changing the basic ridge-style optimization.

Proof sketch of the main bound. Use the decomposition

$$JJ^\top - KK^\top = J(J-K)^\top + (J-K)K^\top$$

and $\|PQ^\top\|_F \le \|P\|_F\|Q\|_F$. Then set $J = T_c$, $K = Z$, and bound $\|Z\|_F \le \|T_c\|_F + \|T_c - Z\|_F$.

Effect of post-hoc rank truncation on Gram geometry. Let $Z := Y_c\widehat{B}$ and $Z_{(m)} := Y_c\widehat{B}_{(m)}$, and define $\delta := \|Z - Z_{(m)}\|_F$. Then

$$\|ZZ^\top - Z_{(m)}Z_{(m)}^\top\|_F \le (\|Z\|_F + \|Z_{(m)}\|_F)\delta \le (2\|Z\|_F + \delta)\delta,$$

and

$$\delta = \|Y_c(\widehat{B} - \widehat{B}_{(m)})\|_F \le \|Y_c\|_{\mathrm{op}}\|\widehat{B} - \widehat{B}_{(m)}\|_F.$$

Evaluating centered Gram mismatch without $n \times n$ matrices. Let $G_S = S_cS_c^\top$ and $G_Z = ZZ^\top$. Then

$$\|G_S - G_Z\|_F^2 = \|S_c^\top S_c\|_F^2 + \|Z^\top Z\|_F^2 - 2\|S_c^\top Z\|_F^2.$$

This identity allows inexpensive evaluation of Gram mismatch using only low-dimensional cross-products.

E.2 Cross-product refinement of the main bound

The operator-norm bound in (4) can be tightened further by exploiting the structure of the residual relative to the target subspace.

Let $E := T_c - Z = T_c - Y_cB$ and $\varepsilon := \|E\|_F$. The decomposition

$$T_cT_c^\top - ZZ^\top \;=\; T_cE^\top + ET_c^\top - EE^\top$$

gives, by the triangle inequality,

$$\|T_cT_c^\top - ZZ^\top\|_F \;\le\; 2\,\|T_c^\top E\|_F + \varepsilon^2. \tag{6}$$

The first term follows from unitary invariance of the Frobenius norm: $\|T_cE^\top\|_F = \|T_c^\top E\|_F$, applied without any norm inequality on T_c.

Relation to earlier bounds. Since $\|T_c^\top E\|_F \le \|T_c\|_{\mathrm{op}}\,\|E\|_F$, (6) implies the operator-norm bound (4); since $\|T_c\|_{\mathrm{op}} \le \|T_c\|_F$, it also implies the Frobenius bound (1). The hierarchy is therefore

$$2\,\|T_c^\top E\|_F + \varepsilon^2 \;\le\; 2\,\|T_c\|_{\mathrm{op}}\,\varepsilon + \varepsilon^2 \;\le\; 2\,\|T_c\|_F\,\varepsilon + \varepsilon^2.$$

When the refinement helps. The quantity $\|T_c^\top E\|_F$ measures how much the residual correlates with the dominant target directions. Ridge regression tends to fit high-variance target directions first, leaving residuals concentrated in the low-variance tail of T_c. When E is approximately orthogonal to the leading singular subspace of T_c, $\|T_c^\top E\|_F$ can be substantially smaller than $\|T_c\|_{\mathrm{op}}\,\varepsilon$, and the bound tightens accordingly.

Practical computation. The cross-product $T_c^\top E \in \mathbb{R}^{q\times q}$ is computed from low-dimensional factors:

$$T_c^\top E = T_c^\top T_c - T_c^\top Z,$$

where $T_c^\top T_c \in \mathbb{R}^{q\times q}$ and $T_c^\top Z \in \mathbb{R}^{q\times q}$ are already available or cheap to form. No $n \times n$ matrix is required, so (6) serves as a computable diagnostic alongside the bound itself.

E.3 Normalized forms and connection to R^2

All three bounds admit a normalized form that connects Gram mismatch to the total coefficient of determination. Let

$$Z := \widehat{T}_c = Y_c \widehat{B}, \qquad E := T_c - Z, \qquad \varepsilon := \|E\|_F.$$

Assume $\|T_c\|_F > 0$, and define

$$\rho := \frac{\varepsilon}{\|T_c\|_F} = \sqrt{1 - R^2_{\text{total}}}, \qquad R^2_{\text{total}} := 1 - \frac{\varepsilon^2}{\|T_c\|_F^2}.$$

Frobenius bound (main text). Dividing (1) by $\|T_c\|_F^2$ gives

$$\frac{\|G_T - G_Z\|_F}{\|T_c\|_F^2} \;\le\; 2\rho + \rho^2.$$

Operator-norm bound. Define the spectral concentration ratio

$$\kappa := \frac{\|T_c\|_{\text{op}}}{\|T_c\|_F}, \qquad \frac{1}{\sqrt{\text{rank}(T_c)}} \;\le\; \kappa \;\le\; 1.$$

The ratio κ is close to 1 when a single target direction dominates and close to $1/\sqrt{\text{rank}(T_c)}$ when target energy is spread evenly across the nonzero singular directions. Dividing (4) by $\|T_c\|_F^2$ gives

$$\frac{\|G_T - G_Z\|_F}{\|T_c\|_F^2} \;\le\; 2\kappa\rho + \rho^2.$$

Cross-product bound. If $\varepsilon > 0$, define the residual–target alignment ratio

$$\gamma := \frac{\|T_c^\top E\|_F}{\|T_c\|_F\, \varepsilon}.$$

If $\varepsilon = 0$, set $\gamma := 0$ by convention. Then

$$0 \;\le\; \gamma \;\le\; \kappa \;\le\; 1.$$

The ratio γ measures how strongly the regression residual correlates with the dominant target directions. Ridge regression tends to leave residuals in the low-variance tail of T_c, driving γ well below κ. Dividing (6) by $\|T_c\|_F^2$ gives

$$\frac{\|G_T - G_Z\|_F}{\|T_c\|_F^2} \;\le\; 2\gamma\rho + \rho^2.$$

Summary hierarchy. In normalized form, the three bounds are nested:

$$2\gamma\rho + \rho^2 \;\le\; 2\kappa\rho + \rho^2 \;\le\; 2\rho + \rho^2, \tag{7}$$

where $\rho = \sqrt{1 - R^2_{\text{total}}}$ captures predictive fit, κ captures target spectral concentration, and γ captures residual–target alignment. All three coincide in the worst case $\gamma = \kappa = 1$; the gaps between them quantify how much structure in the target matrix and in the residual can be exploited.

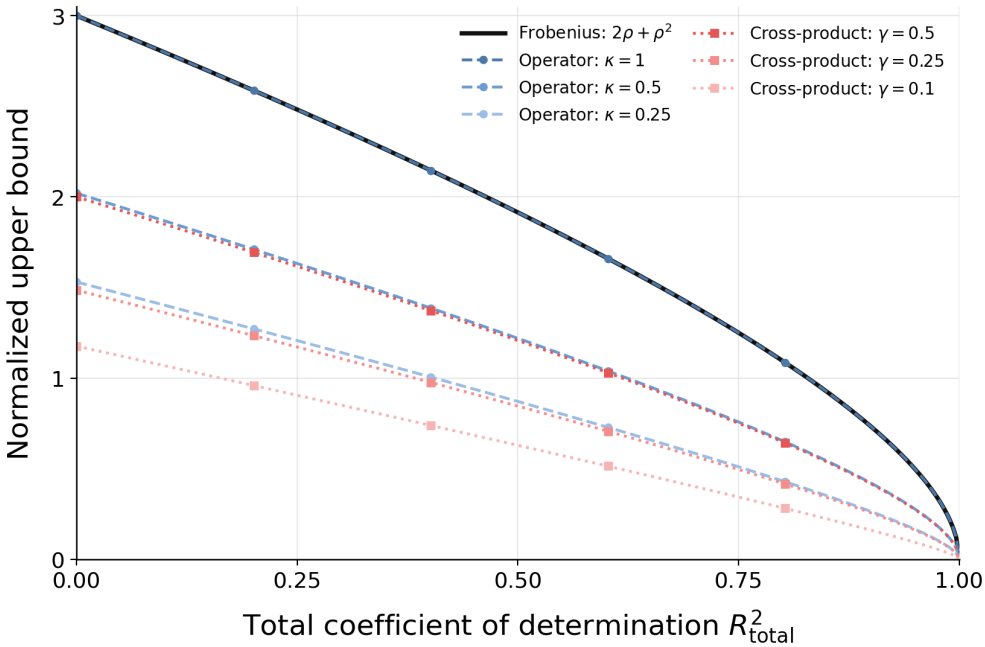

Figure 3: Normalized upper bounds on centered Gram mismatch as a function of total coefficient of determination R^2_{total}. Writing $\rho = \sqrt{1 - R^2_{\text{total}}}$, the universal Frobenius bound is $2\rho + \rho^2$, the operator-norm refinement is $2\kappa\rho + \rho^2$, and the cross-product refinement is $2\gamma\rho + \rho^2$, with $0 \le \gamma \le \kappa \le 1$. The figure shows representative values of κ and γ, using line styles and markers chosen to remain distinguishable in grayscale. All curves decrease monotonically to zero as $R^2_{\text{total}} \to 1$, and for fixed R^2_{total}, smaller κ and γ yield tighter guarantees. Minor visual offsets, if present, are used only to separate nearly overlapping curves and have no mathematical meaning.

Practical note. All three quantities ρ, κ, and γ are computable from low-dimensional cross-products ($q \times q$ matrices) without forming any $n \times n$ object, so they serve as inexpensive diagnostics for assessing both predictive and geometric fit quality.

Appendix F Modular Metric Composition

F.1 Exact composition under a shared ridge operator

Let

$$P_\lambda := (Y_c^\top Y_c + \lambda I_p)^{-1} Y_c^\top$$

denote the ridge operator, and let

$$T_c = [T_{c,1} \mid \cdots \mid T_{c,q}]$$

be the centered target matrix with q scalar columns. Because ridge regression is linear in the target,

$$\widehat{B} = P_\lambda T_c = \left[\, P_\lambda T_{c,1} \;\middle|\; \cdots \;\middle|\; P_\lambda T_{c,q} \,\right] =: \left[\, \widehat{B}_1 \;\middle|\; \cdots \;\middle|\; \widehat{B}_q \,\right],$$

so the induced full metric decomposes additively over targets:

$$\widehat{M} = \widehat{B}\,\widehat{B}^\top = \sum_{j=1}^{q} \widehat{B}_j\,\widehat{B}_j^\top. \tag{8}$$

Proposition 1 (Incremental metric update)**.** *Given a current full metric $\widehat{M}$ built from targets $T_{c,1}, \ldots, T_{c,q}$, a new scalar target $T_{c,q+1}$ can be incorporated by the rank-one update*

$$\widehat{M}' = \widehat{M} + \widehat{B}_{q+1}\widehat{B}_{q+1}^\top, \qquad \widehat{B}_{q+1} = P_\lambda T_{c,q+1},$$

and an existing target $T_{c,j}$ removed by the corresponding rank-one downdate

$$\widehat{M}' = \widehat{M} - \widehat{B}_j\widehat{B}_j^\top.$$

In both cases $\widehat{M}'$ is identical to the full metric obtained by fitting jointly on the modified target set. In particular, $\widehat{M}' \succeq 0$ in both the update and downdate cases, since it remains the jointly fitted metric on the retained target columns.

Proof. Immediate from (8) and the linearity of P_λ. □

Remark 1 (Weighted composition)**.** *More generally, one may define a task-specific metric*

$$\widehat{M}_w = \sum_{j=1}^{q} w_j\, \widehat{B}_j\, \widehat{B}_j^\top, \qquad w_j \geq 0,$$

which is equivalent to fitting on the rescaled centered targets

$$[\sqrt{w_1}\, T_{c,1} \mid \cdots \mid \sqrt{w_q}\, T_{c,q}].$$

The weights $\{w_j\}$ provide a continuous, interpretable control over the relative influence of each target on the induced notion of similarity.

Remark 2 (Simple workflow)**.** *The ridge operator P_λ is computed once. Each new target requires only a single application of the precomputed operator P_λ to the scalar target column $T_{c,j}$, together with storage of the resulting $\widehat{B}_j$. Metric recomposition is then $O(p^2)$ per added or removed target. If a rank-m truncated metric is used downstream, one may compute that truncation either from $\widehat{M}$ or, equivalently, from the singular value decomposition of $\widehat{B} \in \mathbb{R}^{p\times q}$. When $q \ll p$, the latter costs $O(pq^2)$ and avoids a full dense eigendecomposition of $\widehat{M}$; a full eigendecomposition of $\widehat{M}$ is $O(p^3)$ in general.*

F.2 Composition across heterogeneous supervision blocks

The same linear construction extends naturally to supervision blocks observed on different subsets of rows, provided they are all expressed in the same ambient feature coordinates.

Proposition 2 (Composition across subset-specific supervision blocks)**.** *Let $Y_c = HY \in \mathbb{R}^{n\times p}$ be the globally centered ambient feature matrix. For block s, let $I_s \subseteq \{1, \ldots, n\}$ be an index set of size n_s, let $R_s \in \{0,1\}^{n_s\times n}$ be the corresponding row-selection matrix, and define*

$$Y_c^{(s)} := R_s Y_c \in \mathbb{R}^{n_s\times p}.$$

Let

$$T_c^{(s)} \in \mathbb{R}^{n_s\times q_s}$$

denote the corresponding centered target block. Thus, for exact composition in a shared ambient embedding, feature centering is inherited from the globally centered matrix Y_c, while each target block may be centered according to the local fit.

Define the block-specific ridge operator

$$P_\lambda^{(s)} := \big((Y_c^{(s)})^\top Y_c^{(s)} + \lambda I_p\big)^{-1}(Y_c^{(s)})^\top,$$

and the coefficient matrix

$$\widehat{B}^{(s)} := P_\lambda^{(s)} T_c^{(s)}.$$

Each block then contributes the PSD feature-space metric

$$\widehat{M}^{(s)} := \widehat{B}^{(s)}(\widehat{B}^{(s)})^\top \in \mathbb{R}^{p\times p}.$$

For any nonnegative weights $\omega_s \geq 0$, the composite metric

$$\widehat{M}_{\text{het}} := \sum_s \omega_s \widehat{M}^{(s)}$$

is again PSD. Moreover, if one defines the concatenated factor

$$\widetilde{B} := \big[\, \sqrt{\omega_1}\,\widehat{B}^{(1)} \;\big|\; \cdots \;\big|\; \sqrt{\omega_S}\,\widehat{B}^{(S)} \,\big],$$

then

$$\widehat{M}_{\text{het}} = \widetilde{B}\widetilde{B}^\top.$$

Accordingly, the composite geometry admits the explicit ambient embedding

$$\widetilde{Z} := Y_c\widetilde{B},$$

optionally followed by post-hoc rank truncation for a more compact representation.

Proof. By construction,

$$\widetilde{B}\widetilde{B}^\top = \sum_s \omega_s\, \widehat{B}^{(s)}(\widehat{B}^{(s)})^\top = \sum_s \omega_s\, \widehat{M}^{(s)} = \widehat{M}_{\text{het}}.$$

Since each $\widehat{M}^{(s)}$ is PSD and the weights are nonnegative, $\widehat{M}_{\text{het}}$ is also PSD. □

The shared-operator result of the previous subsection is the exact joint-fit specialization. The present formulation broadens the same linear mechanism to separately estimated supervision blocks in a common ambient feature space.

Three practical consequences are especially valuable.

(i) **Sparse but high-value supervision can steer global geometry.** A supervision block need not be observed on all n rows in order to influence the metric. Small but strategically valuable target blocks—for example, expert annotations, user-provided preference examples, relevance judgments, or rare outcomes—can each contribute a weighted PSD term without forcing all targets into one common n-row matrix and without introducing masked pseudo-targets for unobserved rows.

(ii) **Modular steering and personalization become natural.** A new supervision source contributes its own factor $\widehat{B}^{(s)}$, which can be inserted, revised, reweighted, or removed without refitting unrelated blocks. In retrieval or search over a large embedded corpus, for example, a user may provide only a small number of preference examples; these can define an additional supervision block that steers neighborhoods toward the user's notion of relevance. When the resulting geometry must then be propagated across the full dataset, the principal remaining global operation is the embedding multiply $Y_c\widetilde{B}$.

(iii) **The composed geometry remains explicit, stable, and auditable.** Nonnegative weights preserve positive semidefiniteness automatically, blockwise contributions remain separately inspectable, and the combined factor $\widetilde{B}$ provides a single object from which distances, embeddings, clustering, search, reweighting, and post-hoc truncation can all be derived. Thus heterogeneous supervision sources can be combined in a way that is both modular and geometrically coherent.

The common-n formulation remains the most reusable and computationally efficient specialization, since a single operator P_λ can be applied across all target columns. The subset-specific extension broadens the same linear framework to supervision sources of unequal coverage while preserving exact additivity at the level of PSD feature-space metrics.

Appendix G Optional centered Gram-alignment augmentation

To incorporate an explicit target-geometry preference while controlling embedding rank, one may start from centered reduced-rank ridge regression and add a centered Gram-alignment cross-term:

$$\min_{B\in\mathbb{R}^{p\times q}:\ \mathrm{rank}(B)\le m} \|T_c - Y_cB\|_F^2 + \lambda\|B\|_F^2 - \alpha\,\mathrm{tr}\Big(G_S(Y_cB)(Y_cB)^\top\Big), \qquad \alpha\ge 0,$$

where $G_S := S_cS_c^\top$ for a chosen target factor S. This term is proportional to the usual linear-kernel HSIC quantity and can be implemented through low-rank factors rather than explicit $n\times n$ matrices.[2, 6, 10]

Ignoring the rank constraint, the corresponding normal equations are

$$\Big(Y_c^\top Y_c + \lambda I_p - \alpha\, CC^\top\Big)\,\widehat{B} = Y_c^\top T_c, \qquad C := Y_c^\top S_c.$$

Appendix H Empirical Metrics Reference

Predictive metrics. For continuous targets with residuals $e_i = \hat{y}_i - y_i$ and $\bar{y} = \frac{1}{n}\sum_i y_i$,

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^n e_i^2},$$

$$R^2 = 1 - \frac{\sum_{i=1}^n e_i^2}{\sum_{i=1}^n (y_i-\bar{y})^2}.$$

Purity. If n_{kj} denotes the contingency count between cluster k and label j, then

$$\mathrm{Purity} = \frac{1}{n}\sum_{k=1}^K \max_j n_{kj}.$$

Best-map accuracy (Hungarian assignment).

$$\mathrm{BestMapAcc} = \frac{1}{n}\max_\pi \sum_{i=1}^n \mathbf{1}[\pi(c_i)=\tilde{y}_i]\,,$$

where π is a one-to-one mapping from cluster IDs to label IDs.

Adjusted Rand Index (ARI).

$$\mathrm{ARI} = \frac{\sum_{k,j}\binom{n_{kj}}{2} - \frac{\left(\sum_k \binom{n_{k\cdot}}{2}\right)\left(\sum_j \binom{n_{\cdot j}}{2}\right)}{\binom{n}{2}}}{\frac{1}{2}\left[\sum_k \binom{n_{k\cdot}}{2} + \sum_j \binom{n_{\cdot j}}{2}\right] - \frac{\left(\sum_k \binom{n_{k\cdot}}{2}\right)\left(\sum_j \binom{n_{\cdot j}}{2}\right)}{\binom{n}{2}}}.$$

Normalized Mutual Information (NMI). Let n_{kj} denote the contingency count between cluster k and label j, with $n_{k\cdot} = \sum_j n_{kj}$, $n_{\cdot j} = \sum_k n_{kj}$, and $n = \sum_{k,j} n_{kj}$. Define the empirical mutual information

$$I(C;Y) = \sum_{k,j} \frac{n_{kj}}{n} \log\left(\frac{n\, n_{kj}}{n_{k\cdot} n_{\cdot j}}\right),$$

where terms with $n_{kj} = 0$ are taken as zero. Let

$$H(C) = -\sum_k \frac{n_{k\cdot}}{n} \log\left(\frac{n_{k\cdot}}{n}\right), \qquad H(Y) = -\sum_j \frac{n_{\cdot j}}{n} \log\left(\frac{n_{\cdot j}}{n}\right).$$

A common normalized version is

$$\mathrm{NMI} = \frac{I(C;Y)}{\sqrt{H(C)\,H(Y)}}.$$

Higher values of ARI and NMI indicate stronger agreement between clusters and reference labels, with 1 corresponding to perfect agreement.

Continuous outcomes and binned external metrics. External clustering metrics such as ARI, purity, and best-map accuracy require discrete reference labels. For continuous outcomes, one practical device is to discretize targets into quantile bins using cut points fit on the training split and then applied unchanged to the test split.

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